

Tool Based STQA Activity

AI-ASSISTED ENVIRONMENTAL IMPACT ASSESSMENT (EIA) PLATFORM

Test Scenarios and Test Cases using Black Box Testing Techniques

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(2026-2027)

CERTIFICATE

This is to certify that the STQA Activity

Test Scenario and Test Case Design for the AI-Assisted Environmental Impact Assessment (EIA) Platform using Black Box Testing Techniques and AI Tools

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Abstract

Preparing the baseline for an Environmental Impact Assessment by hand means collecting air quality, weather, climate, land-use, ecology, population and hazard data from many separate sources. The AI-Assisted EIA Platform automates this for an industrial project site in Solapur district, Maharashtra. It queries 13 public data providers at the same time, measures the distance from the site to sensitive receptors, and reports for every data item whether it was found. It never fills a gap with an invented value.

This report documents the black box test design for the platform. It identifies ten functional modules, defines 15 test scenarios and presents 71 detailed test cases (EIA-TC-001 to EIA-TC-071) in the standard test case template. The cases were generated with the help of two AI tools and then reviewed by the project team against the running application. Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, State Transition Testing and Use Case Testing are applied wherever they suit the input rules and workflows of the system. The input rules used are the ones the current build enforces; the few that are assumptions are marked (A).

Coverage is shown by a Requirements Traceability Matrix linking each of the 14 functional requirements to its scenarios and test cases. The impact calculation engine, recommendations and report generation are not implemented yet, so their 4 test cases are marked Blocked. The Actual Result and Status columns are completed by the team during execution, and defects found are logged in the defect register in Chapter 5.

CHAPTER 1: INTRODUCTION

1.1 Project Overview

The AI-Assisted Environmental Impact Assessment (EIA) Platform is a web application that helps environmental consultants and project developers prepare the environmental baseline for an industrial project. The user registers the project and its site (latitude and longitude), and the platform collects the baseline for that point: air quality, meteorology and climate, terrain, soil, land use, ecology, nearby receptors, population and natural hazards.

The platform is decision support only. A large language model (LLM), planned for a later phase, will only explain structured results and will never make an environmental decision. Regulatory limits (NAAQS 2009, Noise Rules 2000, EP Rules Schedule VI, IS 10500) are stored with a verified flag, and an unverified limit is never presented as a compliance statement. The current scope is Solapur district, Maharashtra.

1.2 Core Platform Workflows

1.2.1 User Registration and Authentication

Users register as Project Developer, Environmental Consultant, Sustainability Officer or Administrator and sign in with email and password. Passwords are stored as bcrypt hashes, and a JSON Web Token (JWT) valid for 7 days authorises every API call. Each user can see and change only their own projects and assessments.

1.2.2 Project and Site Registration

A project records its name, industry (13 sectors such as Sugar, Textile, Cement and Distillery), type, land area, investment, employees and operating hours. Its site records latitude, longitude, city, district, state, an area classification (Industrial, Commercial, Residential, Silence Zone or Rural/Other) and whether it lies in an ecologically sensitive area.

1.2.3 Assessments and Assessment Inputs

A project can have several assessments, numbered 1, 2, 3 and so on. Values that cannot be derived from a coordinate, such as measured noise levels, fuel consumption or waste generation, are entered as assessment inputs in 12 categories: Air, Water, Ecology, Carbon, Resource, Waste, Noise, Meteorology, Land, Soil, Socio-economic and Natural Hazards.

1.2.4 Environmental Baseline Collection

For a search radius of up to 25 km the analytics service queries 13 providers concurrently: OpenAQ, CPCB real-time AQI (data.gov.in), six Open-Meteo APIs (air quality, weather, historical climate, climate projections, elevation and flood), OpenStreetMap Overpass, two SoilGrids APIs, WorldPop and the USGS earthquake catalogue. Each provider reports available, unavailable, error or skipped. Readings older than 30 days are dropped, and CPCB values are kept as AQI sub-indices, separate from concentrations.

1.2.5 GIS Proximity Analysis and Data Coverage

The platform finds the nearest school, hospital, settlement, road, railway, airport, river, lake, wetland, forest, industrial area and protected area, with the distance in metres and whether the site lies inside the

feature. When nothing is found it says "none found within X km". A data catalogue reports the status of every item in the specification and lists the inputs the user still has to provide.

1.2.6 Reference Data and Dashboard

Users can browse regulatory standards, engineering coefficients and calculation rules, filtered by category, zone, industry or factor. The dashboard shows project and assessment counts, the number of regulatory limits not yet verified and the number of usable coefficients.

1.3 Scope of Black Box Testing

Testing covers the functional behaviour of the platform as seen through the React user interface and the REST APIs, without examining source code. It covers registration, login and sessions, token handling and data isolation, project and site validation, assessments and inputs, the baseline fetch, provider status reporting, air-quality data rules, GIS proximity, the data catalogue, reference data, the dashboard and the stored baseline lifecycle.

The external providers are treated as black boxes. The tests check that the platform responds correctly to what they return, including empty results, failures, timeouts and missing credentials; the accuracy of the providers' own data is not judged. Impact calculation, compliance comparison, LLM explanations and report generation are not built yet, so their cases are Blocked. Load testing, security penetration testing and ML model evaluation are outside the scope of this report.

CHAPTER 2: SYSTEM ARCHITECTURE

2.1 System Architecture Overview

The platform has three services. A React web frontend (Vite, TypeScript, Leaflet maps) calls a Node.js/Express REST backend secured with JWT. The backend owns authentication, projects, sites, assessments and all database persistence in PostgreSQL with PostGIS. For the baseline it calls a separate, stateless Python FastAPI analytics service, which queries the external providers, normalises their responses, computes GIS distances and returns structured JSON; it never writes to the database. The three services and the database run together in a Docker Compose development stack.

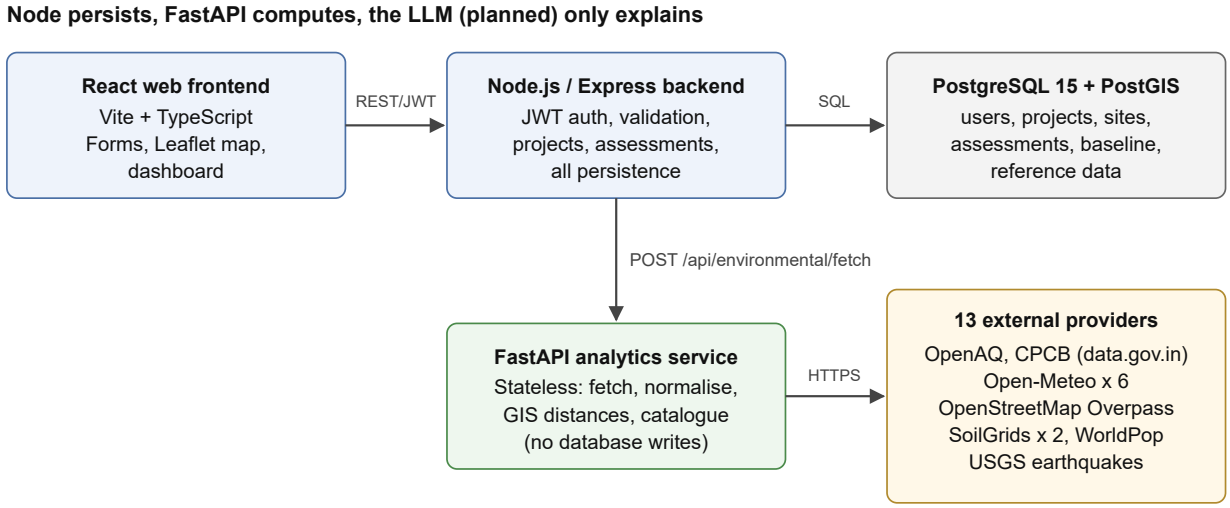


Figure 2.1: System Architecture

2.2 Major Functional Modules

The platform is divided into ten functional modules for testing purposes.

Table 2.1: Major Functional Modules

ID	Module	Description
M1	Authentication and Session Management	Registration, login, logout, JWT sessions, deactivated accounts, session expiry in the UI
M2	Token Validation and Data Isolation	Rejects missing, tampered or expired tokens; limits every project, assessment and dataset to its owner
M3	Project and Site Management	Create, list, search, update and delete projects with their site location and area classification
M4	Assessment Management and Inputs	Numbered assessments per project; assessment inputs in 12 categories
M5	Environmental Baseline Collection	Fetch request validation, concurrent calls to 13 providers, provider outcomes, storing and replacing results
M6	Air-Quality Data Processing	OpenAQ, CPCB and Open-Meteo readings; station selection, stale and NA readings, AQI sub-indices
M7	GIS Proximity and Population Analysis	Nearest receptors and features from OpenStreetMap, inside/outside tests, population within 1, 5 and 10 km
M8	Data Catalogue and Required Inputs	Status of every specification item; required inputs and whether they are provided
M9	Reference Data and Dashboard	Regulatory standards, coefficients and rules with verification status; dashboard summary
M10	Impact Calculation, Recommendations and Reports	Impact results, compliance comparison, LLM explanation and report generation (not yet implemented in the build)

2.3 End-to-End System Processing Flow

A user signs in and creates a project with its site. They start an assessment and enter the values that cannot come from a coordinate. When they fetch the baseline, the backend validates the request, checks ownership and the site coordinates, and sends the site, project and inputs to the analytics service. The service queries the 13 providers concurrently, each with a 240-second limit, and returns observations, GIS features, the catalogue status and every provider's outcome. The backend stores the results in one transaction, replacing earlier rows from the same providers, and the UI shows provider outcomes, observations and GIS results. Impact calculation, recommendations and reports will follow once the calculation engine is built.

CHAPTER 3: TESTING ENVIRONMENT & REQUIREMENTS

3.1 Functional Testing Requirements

The following functional requirements (FR) were identified from the project specification and the current build, and are used for traceability.

Table 3.1: Functional Requirements

FR ID	Requirement	Module
FR-01	The system shall let users register and sign in and out using JWT-based authentication	M1
FR-02	The system shall reject missing, malformed, tampered or expired tokens and restrict every project, assessment and dataset to its owner	M2
FR-03	The system shall create a project with valid details and its site location	M3
FR-04	The system shall validate site coordinates, area classification and numeric project fields	M3
FR-05	The system shall let users list, search, update and delete their own projects	M3
FR-06	The system shall create sequentially numbered assessments and record inputs in valid categories	M4
FR-07	The system shall collect the environmental baseline for the site within a valid search radius and store it	M5
FR-08	The system shall report every provider's outcome (available, unavailable, error, skipped) and continue when a provider fails	M5
FR-09	The system shall drop stale and invalid air-quality readings and keep CPCB AQI sub-indices separate from concentrations	M6
FR-10	The system shall compute distances to sensitive receptors and population around the site, and report absence explicitly	M7
FR-11	The system shall report the status of every data catalogue item and which required inputs have been provided	M8
FR-12	The system shall show regulatory standards and coefficients with their verification status, and a summary dashboard	M9
FR-13	The system shall keep the stored baseline consistent across re-fetches and failed fetches	M5
FR-14	The system shall calculate impacts, compare compliance, give recommendations and generate reports (not implemented in the current build)	M10

3.2 Non-Functional Testing Constraints

- **Performance:** a baseline fetch must finish within the backend's 270-second limit to the analytics service (the UI waits up to 300 seconds); each provider is given at most 240 seconds. Other API calls should answer within the UI's 30-second limit.
- **Security:** protected pages and APIs must reject missing, malformed, tampered or expired tokens; passwords are stored only as bcrypt hashes; provider error messages must never contain API keys.
- **Usability:** the main path (create project, start assessment, fetch baseline) should be reachable in a few steps, and every error should be shown as a readable message, not a raw status code.
- **Reliability:** one failing provider must not fail the others, and a failed fetch must not leave partial data. Bad input must produce a clear message and must not corrupt data.

- **Data integrity:** no value may be invented; modelled, observed and derived values are labelled; unverified regulatory limits are reported as unverified.

Non-functional items are checked only through observable behaviour in this activity. They are not part of the pass/fail count unless a test case says so.

3.3 Software and Hardware Testing Environment Baseline

Versions below are taken from the project's package files; confirm the browser and hardware details before execution.

Table 3.2: Testing Environment

Parameter	Specification
Client	React 19 web app (Vite 8, TypeScript 6, Tailwind CSS 4, Leaflet 1.9) on a current browser [confirm browser and version]
Security layer	JWT (jsonwebtoken 9, 7-day expiry), bcrypt password hashing, helmet headers, express-validator request validation
Backend	Node.js 24 with Express 4: auth, projects, assessments, inputs, baseline storage, reference data
Analytics service	Python 3.10 with FastAPI 0.115, httpx and Pydantic 2: provider clients, mappers, GIS distances, data catalogue
Database	PostgreSQL 15 with PostGIS 3.4
Deployment	Docker Compose development stack (frontend, backend, analytics service, database)
External providers	OpenAQ v3 (API key), CPCB via data.gov.in (OGD API key), Open-Meteo, OpenStreetMap Overpass, SoilGrids, WorldPop, USGS
API testing	Postman or curl for API-level cases
Test hardware	[Laptop model, RAM, OS]
Test management	Standard test case template (Excel/Word); optional defect log in JIRA
AI tools used	Claude and [second tool, e.g. ChatGPT or Copilot]

3.4 Test Management Tooling and AI Tools

Test cases are recorded in the standard test case template given in the activity notice (Sr. No., Test Case ID, Test Scenario, Test Case Description, Preconditions, Test Data, Test Steps, Expected Result, Actual Result and Status).

At least two AI tools were used to generate test cases. The same prompt was given to each tool, describing the project, its modules and its input rules. The outputs were compared, merged and checked by the team against the actual application. Section 5.7 records the comparison. AI output is treated as a draft: every case was reviewed, and rules marked (A) are assumptions to be confirmed with the project team.

3.5 Test Data Sets and Initial Preconditions

Table 3.3: Test Data Sets

Data set	Contents
User accounts	One account per role (e.g. consultant1@eia.test); a second user B with its own project; one deactivated account (is_active = FALSE); an expired token (JWT_EXPIRES_IN=1m); a tampered token
Passwords	7, 8 and 9 characters; an 80-character password for the 72-byte boundary
Sites	Solapur city (17.6599, 75.9064); a point inside an OSM industrial-area polygon; a point with no protected area within 1 km; a project with no location (created through the API)
Coordinate boundaries	Latitude -90.0001, -90, 90, 90.0001; longitude -180.0001, -180, 180, 180.0001
Search radius	0, 0.1, 10, 25, 25.1 km, 'ten', and no value
Assessment inputs	Noise leq_day 68 dB(A); Carbon diesel_consumption 120 kL/year; category 'Climate' and 'air'; empty array; blank value
Provider conditions	OPENAQ_API_KEY removed; invalid OGD_API_KEY; unreachable OVERPASS_ENDPOINTS; PROVIDER_DEADLINE_SECONDS=5; analytics service stopped
Stubbed provider data	Stations at 9.9, 10 and 10.1 km; stations last reporting 29, 30 and 31 days ago; a CPCB pollutant value 'NA' (served by a mock server)
Identifiers	'abc' (not a UUID); a random UUID; IDs belonging to user A

Unlike a design built on assumed limits, the rules in Table 3.4 are the ones the current build enforces, with the layer that enforces each. Rules marked (A) are assumptions about the intended behaviour where the build does not define it.

Table 3.4: Implemented Input Rules

Rule	Value	Enforced in
Password length	At least 8 characters; no maximum	UI, API
Email	Valid format; unique (A: case-insensitive)	API
Name	Required; whitespace trimmed	API
Role	One of 4 roles; default Project Developer	API
Token	Bearer JWT, expires after 7 days	API
Project name, industry	Required	UI, API
Latitude / longitude	-90 to 90 / -180 to 180; required in the UI	UI, API, DB
Area classification	Industrial, Commercial, Residential, Silence Zone, Rural/Other	API, DB
Operating hours; employees, land area, capacity	0 to 24; 0 or more (A: rejected with 400)	UI, DB
Project status	Draft, In Progress, Completed, Archived (A: others rejected with 400)	DB
Assessment inputs	At least 1; 12 categories, case-sensitive; parameter name required	API
Search radius	Greater than 0 and at most 25 km; default 25 (API), 10 (UI)	UI, API, service
Provider time limit	240 s each; backend waits 270 s; UI waits 300 s	Service, API, UI
Observation age	Readings older than 30 days dropped	Service
Monitoring stations	At most 3 OpenAQ locations and 3 CPCB stations within the radius, nearest first	Service
OSM search distances	Schools, hospitals, roads 5 km; residential 2 km; settlements, railways, rivers, lakes, wetlands, forests, industrial areas 10 km; airports, towns, protected areas 25 km; each capped at the radius	Service
Population buffers	1, 5 and 10 km (WorldPop 2020)	Service
Earthquakes; climate	Within 300 km, magnitude 4.0+, since 1970; last 10 full years and projections to 2049	Service

CHAPTER 4: BLACK BOX TESTING TECHNIQUES APPLICATION

Five black box techniques are applied. Each is used where it fits the input rule or workflow being tested.

4.1 Equivalence Partitioning (EP)

Inputs are divided into valid and invalid classes, and one value from each class is tested.

Table 4.1: Equivalence Classes

Input	Valid classes	Invalid classes
Login credentials	Registered, active email with correct password	Wrong password; unregistered email; deactivated account; blank fields
Email	Well-formed, not yet registered	Malformed; already registered (any letter case)
Role	One of the 4 roles; omitted (default)	Any other value
Authorization header	'Bearer' + valid, unexpired token	Missing; wrong scheme; tampered; expired
Resource ID	UUID owned by the user	Another user's UUID; unknown UUID; not a UUID
Area classification	One of the 5 classes	Any other value
Input category	One of the 12 categories (exact case)	Unknown category; wrong letter case
Search radius	Number greater than 0 and at most 25; omitted	0 or less; above 25; non-numeric
Provider condition	Configured and responding	Missing key; invalid key; unreachable; slower than the limit
Station reading	Numeric, at most 30 days old	'NA'; negative; older than 30 days

4.2 Boundary Value Analysis (BVA)

Values at and around the edge of each range are tested, because defects often occur there.

Table 4.2: Boundary Values

Input	Range	Test values	Expected
Password length	8 or more	7, 8, 9	7 rejected
Password significance	Whole password	Same first 72 characters, different tail	Different password rejected
Latitude	-90 to 90	-90.0001, -90, 90, 90.0001	Outer values rejected
Longitude	-180 to 180	-180.0001, -180, 180, 180.0001	Outer values rejected
Operating hours	0 to 24	0, 24, 24.1	24.1 rejected
Employees, land area	0 or more	-1, -5	Rejected
Search radius	Above 0, at most 25 km	0, 0.1, 25, 25.1	0 and 25.1 rejected
Station distance	At most the radius (10 km)	9.9, 10, 10.1 km	10.1 km excluded
Reading age	At most 30 days	29, 30, 31 days	31 days excluded
Distance to an area feature	0 when inside	Point inside a polygon	0 m, inside = true

4.3 Decision Table Testing (DTT)

Decision tables cover combinations of conditions. Three are used: the baseline fetch request, provider outcomes and data access. In Table 4.3 the backend checks the conditions in the order listed, so a dash (-) means the condition is not reached.

Table 4.3: Decision Table for the Baseline Fetch Request

Condition / Action	R1	R2	R3	R4	R5	R6
C1: Valid token	T	T	T	T	T	F
C2: Radius valid or omitted	T	T	T	T	F	-
C3: Assessment owned by user	T	T	T	F	-	-
C4: Site has coordinates	T	T	F	-	-	-
C5: Analytics service reachable	T	F	-	-	-	-
Action	201: baseline stored	502: unreachable	422: no coordinates	404: not found	400: radius error	401: token error

Table 4.4: Decision Table for a Provider's Outcome

Condition / Action	R1	R2	R3	R4
C1: Credentials configured (where needed)	T	T	T	F
C2: Call completed within 240 s without error	T	T	F	-
C3: Usable records returned	T	F	-	-
Status reported	available	unavailable	error	skipped

Table 4.5: Decision Table for Data Access

Function	Owner	Other signed-in user	No / invalid token
View, update or delete a project	Allowed	404 Not found	401
Create or list assessments	Allowed	404 Not found	401
Add or view inputs; fetch or view baseline	Allowed	404 Not found	401
View reference data (standards, coefficients, rules)	Allowed	Allowed	401

All four roles currently have the same permissions; access depends on ownership only. If role-based permissions are added, Table 4.5 gains a column per role.

4.4 State Transition Testing (STT)

The user session, a project's status and an assessment's stored baseline move through defined states. Valid and invalid transitions are both tested.

Table 4.6: Session, Project and Baseline State Transitions

From state	Event	To state	Valid?
Signed out	Valid login	Signed in (token stored)	Yes
Signed in	Page reload with a valid token	Signed in	Yes
Signed in	Any request after the token expires	Signed out, with a session-expired message	Yes
Signed in	Sign out	Signed out (token removed)	Yes
Project: Draft	Status set to In Progress	In Progress	Yes
Project: In Progress	Status set to Completed	Completed	Yes
Project: Completed	Status set to Archived	Archived	Yes
Project: any	Status set to an undefined value ('Deleted')	-	No: blocked
Baseline: none stored	Fetch succeeds	Stored	Yes
Baseline: stored	Fetch succeeds again	Stored (same providers' rows replaced)	Yes
Baseline: stored	Provider fails during a re-fetch	Stored without that provider's earlier rows	Yes
Baseline: stored	Fetch fails (502)	Stored, unchanged	Yes

4.5 Use Case Testing (UCT)

Table 4.7: Use Cases

Use case	Actor	Main flow	Scenario
UC-1 Register and sign in	Environmental Consultant	Create account, sign in, reach the dashboard, sign out	TS_01, TS_02
UC-2 Register a project and its site	Project Developer	Enter project details and site, view it on the map, update or delete it	TS_04, TS_05
UC-3 Collect the baseline	Environmental Consultant	Start an assessment, enter inputs, choose a radius, fetch the baseline	TS_06, TS_07, TS_08
UC-4 Review baseline results	Environmental Consultant	Review provider outcomes, air quality, GIS proximity and data coverage	TS_09, TS_10, TS_11
UC-5 Review reference limits	Sustainability Officer	Open the dashboard and reference data, check unverified limits	TS_12
UC-6 Full baseline assessment	All roles	Register, create project, start assessment, enter inputs, fetch, review	TS_15

CHAPTER 5: TEST SUITE EXECUTION AND SUMMARY MATRIX

5.1 Test Scenario Catalog

15 test scenarios cover the major functions of the platform.

Table 5.1: Test Scenario Catalog

Scenario ID	Test scenario	Module	Main techniques	No. of test cases
TS_01	User registration validation	M1	EP, BVA	6
TS_02	User login and session management	M1	EP, BVA, ST	7
TS_03	Token validation and data isolation	M2	DT, EP, ST	7
TS_04	Project creation and site validation	M3	EP, BVA	8
TS_05	Project list, update and delete	M3	EP, ST, UC	4
TS_06	Assessment creation and inputs	M4	EP, DT	5
TS_07	Baseline fetch request validation	M5	DT, BVA, UC	5
TS_08	Provider status reporting and fault tolerance	M5	DT, EP	5
TS_09	Air-quality data quality rules	M6	EP, BVA	4
TS_10	GIS proximity and population analysis	M7	EP, BVA	4
TS_11	Data catalogue and required inputs	M8	EP, DT	3
TS_12	Reference data and dashboard	M9	EP, UC	3
TS_13	Stored baseline lifecycle	M5	ST	3
TS_14	Impact calculation, recommendations and reports (Blocked: not implemented)	M10	UC, EP	5
TS_15	End-to-end baseline assessment workflow	All	UC	2

5.2 Comprehensive Black Box Test Cases Matrix (EIA-TC-001 to EIA-TC-071)

Cases use the standard template from the activity notice. Rules marked (A) are assumptions. The Actual Result column is left blank for the team to fill in during execution, and Status is updated to Pass, Fail or Blocked.

Sr. No.	Test Case ID	Test Scenario	Test Case Description	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status
1	EIA-TC-001	TS_01 - User registration validation	Verify registration with valid details	Application online; email not yet registered	Name: Asha Patil Email: consultant1@eia.test Password: Consult@123 Role: Environmental consultant Organisation: WIT	1. Open the Sign in page and click 'Create one' 2. Enter the data 3. Click Create account	Account created (HTTP 201); JWT issued; response has no password hash; dashboard opens with 'Welcome back, Asha'		
2	EIA-TC-002	TS_01 - User registration validation	Verify password length lower boundary (minimum 8 characters)	Registration form open	Passwords of 7, 8 and 9 characters: Abc@123, Abcd@123, Abcde@123 (new email each)	1. Fill valid details with each password in turn 2. Submit in the UI 3. Repeat the 7-character case with POST /api/auth/register	7 characters rejected: UI blocks submit, API returns 400 'Password must be at least 8 characters.'; 8 and 9 characters accepted		
3	EIA-TC-003	TS_01 - User registration validation	Verify mandatory field validation on registration	Registration form open	Name: (blank) and ' ' Email: (blank) Password: (blank)	1. Submit the form with each field blank in turn 2. Send POST /api/auth/register with name ' '	UI shows required-field prompt and sends no request; API returns 400 'Name is required.'; no account created		
4	EIA-TC-004	TS_01 - User registration validation	Verify invalid email formats are rejected	Registration form open	consultant1@ consultant1.eia.test @eia.test	1. Submit each email with otherwise valid details (UI and API)	400 'A valid email is required.' for each; no account created		
5	EIA-TC-005	TS_01 - User registration validation	Verify duplicate email registration, including a case variant	consultant1@eia.test already registered	Email 1: consultant1@eia.test Email 2: Consultant1@EIA.test	1. Register with Email 1 2. Register with Email 2	Both rejected with 409 'An account with this email already exists.' (A: email addresses are case-insensitive)		
6	EIA-TC-006	TS_01 - User registration validation	Verify role selection and default role	Application online	Role: (omitted); Role: SUPERUSER; each of the 4 valid roles	1. Send POST /api/auth/register with each role value	Omitted role saved as PROJECT_DEVELOPER; SUPERUSER rejected with 400 'Role must be one of: PROJECT_DEVELOPER, ENVIRONMENTAL_CONSULTANT, SUSTAINABILITY_OFFICER, ADMIN'; 4 valid roles accepted		
7	EIA-TC-007	TS_02 - User login and session management	Verify login with valid credentials	User registered; application online	Email: consultant1@eia.test Password: Consult@123	1. Open the Sign in page 2. Enter email and password 3. Click Sign in	HTTP 200 with JWT; token stored; dashboard opens showing the user's name		

Sr. No.	Test Case ID	Test Scenario	Test Case Description	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status
8	EIA-TC-008	TS_02 - User login and session management	Verify login with a wrong password	User registered	Email: consultant1@eia.test Password: Wrong@1234	1. Enter the data 2. Click Sign in	401 'Invalid email or password.' shown; no token issued; user stays on Sign in page		
9	EIA-TC-009	TS_02 - User login and session management	Verify login with an unregistered email	Application online	Email: nouser@eia.test Password: Consult@123	1. Enter the data 2. Click Sign in	401 'Invalid email or password.' (same message as a wrong password, so the response does not reveal whether the account exists)		
10	EIA-TC-010	TS_02 - User login and session management	Verify login with empty fields	Sign in page open	Email: (blank) Password: (blank)	1. Leave both fields blank and click Sign in 2. Send POST /api/auth/login with password ''	UI shows mandatory-field prompts and sends no request; API returns 400 'Password is required.'		
11	EIA-TC-011	TS_02 - User login and session management	Verify a deactivated account cannot log in	Account exists with is_active = FALSE (set by tester in the database)	Email: inactive@eia.test with its correct password	1. Enter the data 2. Click Sign in	403 'This account has been deactivated.'; no token issued		
12	EIA-TC-012	TS_02 - User login and session management	Verify long-password boundary (72 bytes)	Account registered with an 80-character password P = 72 x 'A' + 'Secret#1'	Login password: 72 x 'A' + 'Wrong#99' (same first 72 characters, different tail)	1. Log in with the data password	Login rejected with 401; only the exact 80-character password is accepted		
13	EIA-TC-013	TS_02 - User login and session management	Verify session persists on reload and ends on sign out	Signed in as consultant1	Browser localStorage key eia.token	1. Reload the dashboard 2. Click Sign out in the header 3. Press the browser Back button	After reload the user stays signed in; after Sign out the Sign in page opens and eia.token is removed; Back does not show project data		
14	EIA-TC-014	TS_03 - Token validation and data isolation	Verify protected API without a valid Authorization header	Backend running	GET /api/projects with (a) no header (b) 'Token abc' (c) 'Bearer' with no token	1. Send each request	401 'Missing or malformed Authorization header.' for each; no data returned		
15	EIA-TC-015	TS_03 - Token validation and data isolation	Verify a tampered JWT is rejected	Valid token obtained at login	Same token with one character of the signature changed	1. Call GET /api/auth/me with the tampered token	401 'Invalid or expired token.'		
16	EIA-TC-016	TS_03 - Token validation and data isolation	Verify behaviour with an expired JWT	Backend started with JWT_EXPIRES_IN= 1m; user signed in and idle for 2 minutes	Expired JWT	1. Call any protected API with the token 2. In the UI click All projects	API returns 401 'Invalid or expired token.'; UI shows 'Your session has expired. Please sign in again.' and redirects to Sign in		
17	EIA-TC-017	TS_03 - Token validation and data isolation	Verify a user cannot read, change or delete another user's project	User A owns project P; signed in as user B	Project ID of P	1. As B send GET, PUT and DELETE /api/projects/{P} 2. As B open /projects/{P}/location in the UI	404 'Project not found.' for each request; P unchanged for A; UI shows an error state		
18	EIA-TC-018	TS_03 - Token validation and data isolation	Verify a user cannot access another user's assessment	User A owns assessment A1; signed in as user B	Assessment ID of A1	1. As B call GET /api/assessments/{A1}, GET and POST /inputs, GET and POST /environmental-data	404 'Assessment not found.' for each; nothing is written to A1		

Sr. No.	Test Case ID	Test Scenario	Test Case Description	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status
19	EIA-TC-019	TS_03 - Token validation and data isolation	Verify each role works only with its own data	One account per role, each with one project	Project Developer, Environmental Consultant, Sustainability Officer, Administrator	1. Sign in as each role 2. Open All projects and create a project	Every role can create and list its own projects and sees no other user's projects (A: roles have no separate permissions in the current build)		
20	EIA-TC-020	TS_03 - Token validation and data isolation	Verify malformed IDs give a clear client error	Signed in	GET /api/projects/abc GET /api/assessments/abc POST /api/assessments/abc/environmental-data	1. Send each request with a valid token	400 or 404 with a clear message for each (the fetch route returns 400 'Assessment id must be a UUID.'); never 500 'Internal server error.'		
21	EIA-TC-021	TS_04 - Project creation and site validation	Verify creating a project with valid details and site	Signed in; New project page open	Name: Solapur Sugar Unit Expansion; Industry: Sugar; Type: Industrial; Area: 25 acres; Investment: 120 Crore; Employees: 350; Hours: 16 Site: Lat 17.6599, Lon 75.9064 (Solapur); District: Solapur; State: Maharashtra; Area class: Industrial	1. Enter the data 2. Click Create project	HTTP 201; Location and GIS page opens with a map marker at the site; project listed in All projects with status Draft		
22	EIA-TC-022	TS_04 - Project creation and site validation	Verify mandatory project fields	New project page open	Project name: (blank); Industry: (not selected)	1. Submit with each field missing 2. Send POST /api/projects with project.name '' and no industry	UI blocks submit; API returns 400 'Project name is required.' / 'Industry is required.'; nothing saved		
23	EIA-TC-023	TS_04 - Project creation and site validation	Verify site coordinates are required in the UI	New project page open	Valid project details; Latitude and Longitude blank	1. Click Create project	Error 'Enter the site latitude and longitude. The environmental baseline is fetched for that point.'; no request sent		
24	EIA-TC-024	TS_04 - Project creation and site validation	Verify latitude boundaries (-90 to 90)	New project page open	Latitude: -90.0001, -90, 90, 90.0001 (Longitude 75.9064)	1. Submit each value in the UI 2. Send the outer values through POST /api/projects	-90.0001 and 90.0001 rejected (UI: 'Latitude must be between -90 and 90.'; API: 400); -90 and 90 accepted		
25	EIA-TC-025	TS_04 - Project creation and site validation	Verify longitude boundaries (-180 to 180)	New project page open	Longitude: -180.0001, -180, 180, 180.0001 (Latitude 17.6599)	1. Submit each value in the UI 2. Send the outer values through POST /api/projects	-180.0001 and 180.0001 rejected (UI: 'Longitude must be between -180 and 180.'; API: 400); -180 and 180 accepted		
26	EIA-TC-026	TS_04 - Project creation and site validation	Verify area classification and ecological-sensitivity values	Signed in; token available	areaClassification: each of Industrial, Commercial, Residential, Silence Zone, Rural/Other; 'Forest' ecologicallySensitive: 'maybe'	1. Send POST /api/projects with each value	5 valid classes accepted; 'Forest' rejected with 400 'areaClassification must be one of: Industrial, Commercial, Residential, Silence Zone, Rural/Other'; 'maybe' rejected with 400		

Sr. No.	Test Case ID	Test Scenario	Test Case Description	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status
27	EIA-TC-027	TS_04 - Project creation and site validation	Verify numeric project field limits	Signed in; token available	operatingHoursPerDay: 0, 24, 24.1 employees: -1 landArea: -5	1. Send POST /api/projects with each value and otherwise valid data	0 and 24 accepted; 24.1, -1 and -5 rejected with 400 and a message naming the field; no project created		
28	EIA-TC-028	TS_04 - Project creation and site validation	Verify 'Use current location' fills the coordinates	New project page open in a browser with location services	Browser location permission: Allow; Block	1. Click Use current location with each permission setting	Allow: latitude and longitude filled with 6 decimal places; Block: 'Could not read your location: ...' shown and fields unchanged		
29	EIA-TC-029	TS_05 - Project list, update and delete	Verify project list shows only own projects and filters by search	User A has 3 projects; user B has 1; signed in as A	Search terms: 'Sugar', 'zzz'	1. Open All projects 2. Search 'Sugar' 3. Search 'zzz'	Only A's 3 projects, newest first; search shows matching projects only; 'zzz' shows 'No project matches that filter'		
30	EIA-TC-030	TS_05 - Project list, update and delete	Verify project update accepts allowed fields only	User A owns project P	PUT body: { name: 'Sugar Unit Phase 2', employees: 400, user_id: '<user B id>' }	1. Send PUT /api/projects/{P} as A 2. Send GET /api/projects/{P}	200; name and employees updated; user_id ignored, so P still belongs to A		
31	EIA-TC-031	TS_05 - Project list, update and delete	Verify project status transitions	Project P in status Draft	status: In Progress, Completed, Archived, then 'Deleted'	1. Send PUT /api/projects/{P} with each status in turn	Draft to In Progress to Completed to Archived accepted; 'Deleted' rejected with 400 listing the allowed statuses (A); status unchanged		
32	EIA-TC-032	TS_05 - Project list, update and delete	Verify deleting a project removes its dependent data	Project P with a site, 1 assessment and stored baseline	Project ID of P and its assessment ID	1. Send DELETE /api/projects/{P} 2. Send GET /api/projects/{P} 3. Send GET /api/assessments/{id}	DELETE returns 204; both GETs return 404; P no longer appears in All projects		
33	EIA-TC-033	TS_06 - Assessment creation and inputs	Verify assessments are numbered sequentially per project	Project P with no assessments	Two create requests	1. Send POST /api/projects/{P}/assessments twice 2. Send GET /api/projects/{P}/assessments	Assessment numbers 1 and 2, both status Draft; list shows number 2 first		
34	EIA-TC-034	TS_06 - Assessment creation and inputs	Verify assessment creation for a project the user does not own	Signed in as user B	Project ID owned by user A; random UUID	1. Send POST /api/projects/{id}/assessments with each ID	404 'Project not found.' for each; no assessment created		
35	EIA-TC-035	TS_06 - Assessment creation and inputs	Verify adding valid assessment inputs	Assessment A1 owned by the user	[{Noise, leq_day, 68, dB(A), Consultant survey}, {Carbon, diesel_consumption, 120, kL/year}]	1. Send POST /api/assessments/{A1}/inputs 2. Send GET /api/assessments/{A1}/inputs	201 with 2 rows; GET lists both, sorted by category		

Sr. No.	Test Case ID	Test Scenario	Test Case Description	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status
36	EIA-TC-036	TS_06 - Assessment creation and inputs	Verify invalid inputs are rejected and nothing partial is saved	Assessment A1 owned by the user	(a) inputs: [] (b) category 'Climate' (c) parameterName ' ' (d) one valid row + one row with category 'Climate'	1. Send POST /api/assessments/{A1}/inputs with each body 2. Send GET .../inputs	(a) 400 'inputs must be a non-empty array.' (b),(d) 400 'category must be one of: Air, Water, ...' (c) 400 'parameterName is required.'; no row from any request is saved		
37	EIA-TC-037	TS_06 - Assessment creation and inputs	Verify every input category value	Assessment A1 owned by the user	One input in each of the 12 categories; category 'air' (lower case)	1. Send POST /api/assessments/{A1}/inputs for each	All 12 categories accepted; 'air' rejected with 400 (category names are case-sensitive)		
38	EIA-TC-038	TS_07 - Baseline fetch request validation	Verify search radius boundaries (greater than 0, at most 25 km)	Assessment for a project with a site	radiusKm: 0, 0.1, 25, 25.1, 'ten'	1. Send POST /api/assessments/{id}/environmental-data with each value 2. Enter 0 and 25.1 in the UI Search radius and click Fetch	0, 25.1 and 'ten' rejected with 400 'radiusKm must be greater than 0 and at most 25.' (UI: 'Radius must be greater than 0 and at most 25 km.'); 0.1 and 25 accepted (201)		
39	EIA-TC-039	TS_07 - Baseline fetch request validation	Verify the default radius when none is sent	Assessment for a project with a site	Request body: {}	1. Send POST /api/assessments/{id}/environmental-data	201; providers' request_parameters show a 25 km radius (radius_m 25000)		
40	EIA-TC-040	TS_07 - Baseline fetch request validation	Verify fetch for a project with no coordinates	Project created through the API without a location; assessment created	Assessment ID	1. Send POST .../environmental-data 2. Open the project's Location and GIS page	422 'The project has no latitude/longitude. Add the site location before fetching environmental data.'; UI shows the no-coordinates message and Fetch is disabled		
41	EIA-TC-041	TS_07 - Baseline fetch request validation	Verify fetch when the analytics service is down	Analytics (FastAPI) service stopped; backend running	Assessment with a site	1. Click Fetch environmental data	502 'Analytics service is unreachable.' shown; button enabled again; previously stored data unchanged		
42	EIA-TC-042	TS_07 - Baseline fetch request validation	Verify the UI during a long baseline fetch	All services running; assessment with a site	Search radius: 10 km	1. Click Fetch environmental data 2. Observe the page until the fetch ends	Radius field and button disabled while running; result within 300 s; provider outcomes and stored observations shown afterwards		
43	EIA-TC-043	TS_08 - Provider status reporting and fault tolerance	Verify every provider reports an outcome	All services and API keys configured	Lat 17.6599, Lon 75.9064 (Solapur); radius 10 km	1. Fetch the baseline 2. Inspect the providers list in the response and UI	13 providers listed, each with status available / unavailable / error / skipped, record count and response time; observations grouped by category		
44	EIA-TC-044	TS_08 - Provider status reporting and fault tolerance	Verify a provider without credentials is skipped	OPENAQ_API_KEY removed and analytics service restarted	Same site and radius	1. Fetch the baseline	openaq status 'skipped' with a reason; other providers unaffected; response still 201		

Sr. No.	Test Case ID	Test Scenario	Test Case Description	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status
45	EIA-TC-045	TS_08 - Provider status reporting and fault tolerance	Verify one failing provider does not fail the others	OVERPASS_ENDPOINTS set to an unreachable host	Same site and radius	1. Fetch the baseline	osm_overpass status 'error' with reason 'All Overpass endpoints failed (...)'; GIS-based catalogue items show error; other providers return data		
46	EIA-TC-046	TS_08 - Provider status reporting and fault tolerance	Verify provider errors never expose API keys	OGD_API_KEY set to an invalid value	Same site and radius	1. Fetch the baseline 2. Read the cpcb_ogd reason, warnings and stored fetch logs	cpcb_ogd status 'error' with a reason such as 'HTTP 403 from api.data.gov.in'; no API key or full URL in any message		
47	EIA-TC-047	TS_08 - Provider status reporting and fault tolerance	Verify the per-provider time limit	PROVIDER_DEADLINE_SECONDS=5; analytics service restarted	Same site and radius	1. Fetch the baseline	Providers slower than 5 s report 'error' with 'No result within 5 s.'; response still lists all 13 providers		
48	EIA-TC-048	TS_09 - Air-quality data quality rules	Verify CPCB values are kept as AQI sub-indices	A CPCB station within the radius is reporting	Site near a CPCB station; radius 25 km	1. Fetch the baseline 2. Inspect CPCB observations	Parameter keys end in _aqi_subindex (e.g. pm25_aqi_subindex), unit 'AQI sub-index'; never stored as pm25 in µg/m³		
49	EIA-TC-049	TS_09 - Air-quality data quality rules	Verify 'NA' pollutant values are skipped	Stubbed CPCB response with one pollutant value 'NA'	Station 'Test Station', PM10 = NA	1. Fetch the baseline	No PM10 row stored; warning 'CPCB station 'Test Station' reported NA for 1 pollutant(s).'		
50	EIA-TC-050	TS_09 - Air-quality data quality rules	Verify stale-reading boundary (30 days)	Stubbed station responses	Stations last reporting 29, 30 and 31 days ago	1. Fetch the baseline	29- and 30-day stations included; 31-day station excluded with warning 'Ignored 1 CPCB station(s) in range that have not reported in the last 30 days.'		
51	EIA-TC-051	TS_09 - Air-quality data quality rules	Verify station radius boundary and station limit	Stubbed CPCB stations; radius 10 km	Stations at 9.9, 10 and 10.1 km, plus 2 more within 5 km	1. Fetch the baseline with radius 10 km	10.1 km station excluded; at most 3 nearest stations kept, sorted by distance		
52	EIA-TC-052	TS_10 - GIS proximity and population analysis	Verify nearest sensitive receptors are reported with distances	Overpass reachable	Lat 17.6599, Lon 75.9064 (Solapur); radius 10 km	1. Fetch the baseline 2. Open the GIS results	Nearest road, railway, school, hospital, settlement, river and lake listed with distance in metres and source OpenStreetMap; school, hospital and settlement counts shown		
53	EIA-TC-053	TS_10 - GIS proximity and population analysis	Verify a site inside an area feature	Overpass reachable	Coordinate inside an OSM industrial-area polygon (e.g. Solapur MIDC; confirm on openstreetmap.org)	1. Create a project at that point 2. Fetch the baseline	Nearest industrial area has distance 0 m and inside_boundary = true		
54	EIA-TC-054	TS_10 - GIS proximity and population analysis	Verify absence is reported explicitly	Overpass reachable	Site with no protected area within 1 km; radius 1 km	1. Fetch the baseline 2. Inspect the protected-area item	Reported as 'none found within 1 km'; item status available; no distance invented		

Sr. No.	Test Case ID	Test Scenario	Test Case Description	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status
55	EIA-TC-055	TS_10 - GIS proximity and population analysis	Verify population within 1, 5 and 10 km	WorldPop reachable	Lat 17.6599, Lon 75.9064 (Solapur)	1. Fetch the baseline 2. Inspect population rows	Population and density reported for 1, 5 and 10 km; population within 1 km <= 5 km <= 10 km		
56	EIA-TC-056	TS_11 - Data catalogue and required inputs	Verify every catalogue item has a status	All services running	Lat 17.6599, Lon 75.9064 (Solapur)	1. Fetch the baseline 2. Inspect the sections in the response	13 sections, every item with a status; surface-water and groundwater items 'unavailable' with a reason that no CPCB/CWC or CGWB dataset is integrated		
57	EIA-TC-057	TS_11 - Data catalogue and required inputs	Verify a required input changes from project_input to available	Assessment with no inputs	Input: Noise, leq_day, 68 dB(A)	1. Fetch the baseline 2. Add the input 3. Fetch again	First fetch: Leq Day 'project_input', reason 'Cannot be derived from latitude/longitude; enter it as an assessment input.', provided = false. Second: 'available', source assessment_input, summary 'Leq Day: 68 dB(A)', provided = true		
58	EIA-TC-058	TS_11 - Data catalogue and required inputs	Verify project land area fills the land requirement and blank inputs are ignored	Project land area 25 acres; no land_requirement input	Input with valueNumeric null and valueText '' for soil_phosphorus	1. Add the blank input 2. Fetch the baseline	Land requirement 'available' with source 'project'; Phosphorus remains 'project_input'		
59	EIA-TC-059	TS_12 - Reference data and dashboard	Verify regulatory standards report their verification status	Reference data seeded	GET /api/reference/standards	1. Send the request 2. Open the dashboard	unverified_count equals the number of rows with verified = false; Regulatory limits card shows the total and 'N not yet checked'		
60	EIA-TC-060	TS_12 - Reference data and dashboard	Verify reference data filters	Reference data seeded	/standards?category=Noise&zone=Industrial /standards?verified=true /coefficients?industry=Sugar	1. Send each request	Only Noise rows for zone Industrial or All; only verified rows; only Sugar or All coefficients		
61	EIA-TC-061	TS_12 - Reference data and dashboard	Verify dashboard counts and empty state	New user with no projects	Then 2 projects with 3 assessments in total	1. Open the dashboard 2. Create the projects and assessments 3. Reload the dashboard	First: 'No projects yet', counts 0. After: Projects = 2, Assessments = 3; recent projects listed		
62	EIA-TC-062	TS_13 - Stored baseline lifecycle	Verify a re-fetch replaces earlier rows	Assessment with one successful fetch	Same site and radius	1. Fetch again 2. Send GET .../environmental-data	Rows from the same providers replaced, not duplicated; fetch_logs show only the latest fetch's 13 outcomes		
63	EIA-TC-063	TS_13 - Stored baseline lifecycle	Verify a provider failing on re-fetch leaves no stale values	First fetch with Overpass available; then OVERPASS_ENDPOINTS set to an unreachable host	Same site and radius	1. Fetch again 2. Inspect GIS results	Earlier OpenStreetMap GIS rows removed; fetch log shows osm_overpass 'error'		

Sr. No.	Test Case ID	Test Scenario	Test Case Description	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status
64	EIA-TC-064	TS_13 - Stored baseline lifecycle	Verify a failed fetch keeps the stored baseline	Assessment with one successful fetch; analytics service then stopped	Same site and radius	1. Fetch again 2. Send GET .../environmental-data	502 returned; observations, GIS results and fetch logs identical to before		
65	EIA-TC-065	TS_14 - Impact calculation, recommendations and reports	Verify impact calculation from assessment inputs	Calculation engine available; verified coefficients loaded	Carbon input diesel_consumption 120 kL/year	1. Run the calculation for the assessment 2. Open Impact results	Result per factor with unit, coefficient and source; missing coefficient shown as not calculated		**Blocked**
66	EIA-TC-066	TS_14 - Impact calculation, recommendations and reports	Verify compliance comparison against verified limits	Calculation engine available; verified NAAQS / noise limits	Input Noise leq_day 78 dB(A) for an Industrial area	1. Run the comparison	Only user-entered monitoring data with an averaging period is compared; unverified limits never reported as compliance		**Blocked**
67	EIA-TC-067	TS_14 - Impact calculation, recommendations and reports	Verify recommendations and AI explanation	Impact results exist; LLM configured	Assessment with results	1. Generate recommendations 2. Open the AI explanation	Recommendations listed by priority; explanation only describes stored results and makes no environmental decision		**Blocked**
68	EIA-TC-068	TS_14 - Impact calculation, recommendations and reports	Verify EIA report generation	Results and recommendations exist	Assessment with results	1. Click Generate report 2. Download the file	PDF report with project, baseline, results, data sources and their statuses		**Blocked**
69	EIA-TC-069	TS_14 - Impact calculation, recommendations and reports	Verify modules not yet built are clearly marked	Signed in	Sidebar: Assessment inputs, Impact results, Reports; URL /reports	1. Hover each sidebar item 2. Type /reports in the address bar	Items greyed out, not clickable, with a 'Soon' badge and a tooltip reason (e.g. 'Available once the calculation engine is built'); /reports redirects to the dashboard		
70	EIA-TC-070	TS_15 - End-to-end baseline assessment workflow	Verify the full baseline workflow	All services running; API keys configured	New consultant account; Sugar project at Lat 17.6599, Lon 75.9064 (Solapur); inputs leq_day 68 dB(A), diesel 120 kL/year	1. Register and sign in 2. Create the project with its site 3. Start an assessment and add the inputs 4. Fetch the baseline at 10 km 5. Review providers, observations, GIS results and catalogue 6. Open the dashboard	Every step succeeds; baseline stored with 13 provider outcomes; required inputs for Leq Day and diesel shown as provided; dashboard counts updated		
71	EIA-TC-071	TS_15 - End-to-end baseline assessment workflow	Verify the workflow with degraded providers	OPENAQ_API_KEY removed; OVERPASS_ENDPOINTS unreachable	Same as previous case	1. Repeat the full workflow 2. Review provider outcomes and observations	Workflow completes; openaq 'skipped', osm_overpass 'error' with reasons; no observation or GIS rows from those providers (no invented values)		

5.3 Formal Requirements Traceability Matrix (RTM)

The matrix links each functional requirement to its scenario and test cases, which shows that every requirement has coverage.

Table 5.2: Requirements Traceability Matrix

FR ID	Requirement	Scenario	Test case IDs
FR-01	Let users register and sign in and out using JWT-based authentication	TS_01, TS_02	EIA-TC-001 to EIA-TC-013
FR-02	Reject missing, malformed, tampered or expired tokens and restrict every project, assessment and dataset to its owner	TS_03	EIA-TC-014 to EIA-TC-020
FR-03	Create a project with valid details and its site location	TS_04	EIA-TC-021 to EIA-TC-023, EIA-TC-028
FR-04	Validate site coordinates, area classification and numeric project fields	TS_04	EIA-TC-024 to EIA-TC-027
FR-05	Let users list, search, update and delete their own projects	TS_05	EIA-TC-029 to EIA-TC-032
FR-06	Create sequentially numbered assessments and record inputs in valid categories	TS_06	EIA-TC-033 to EIA-TC-037
FR-07	Collect the environmental baseline for the site within a valid search radius and store it	TS_07	EIA-TC-038 to EIA-TC-042
FR-08	Report every provider's outcome (available, unavailable, error, skipped) and continue when a provider fails	TS_08	EIA-TC-043 to EIA-TC-047
FR-09	Drop stale and invalid air-quality readings and keep CPCB AQI sub-indices separate from concentrations	TS_09	EIA-TC-048 to EIA-TC-051
FR-10	Compute distances to sensitive receptors and population around the site, and report absence explicitly	TS_10	EIA-TC-052 to EIA-TC-055
FR-11	Report the status of every data catalogue item and which required inputs have been provided	TS_11	EIA-TC-056 to EIA-TC-058
FR-12	Show regulatory standards and coefficients with their verification status, and a summary dashboard	TS_12	EIA-TC-059 to EIA-TC-061
FR-13	Keep the stored baseline consistent across re-fetches and failed fetches	TS_13	EIA-TC-062 to EIA-TC-064
FR-14	Calculate impacts, compare compliance, give recommendations and generate reports (not implemented in the current build)	TS_14	EIA-TC-065 to EIA-TC-069
All	End-to-end workflow across all modules	TS_15	EIA-TC-070, EIA-TC-071

5.4 Test Case to Testing Technique Mapping Profile

Each test case is counted once, under its main technique.

Table 5.3: Technique Mapping

Technique	Test cases	Test case IDs
Equivalence Partitioning (EP)	35	EIA-TC-001, EIA-TC-003 to EIA-TC-011, EIA-TC-015, EIA-TC-020 to EIA-TC-023, EIA-TC-026, EIA-TC-028, EIA-TC-030, EIA-TC-033, EIA-TC-035 to EIA-TC-037, EIA-TC-039, EIA-TC-043, EIA-TC-046, EIA-TC-048, EIA-TC-049, EIA-TC-052, EIA-TC-054 to EIA-TC-056, EIA-TC-058 to EIA-TC-060, EIA-TC-069
Boundary Value Analysis (BVA)	9	EIA-TC-002, EIA-TC-012, EIA-TC-024, EIA-TC-025, EIA-TC-027, EIA-TC-038, EIA-TC-050, EIA-TC-051, EIA-TC-053
Decision Table Testing (DT)	11	EIA-TC-014, EIA-TC-017 to EIA-TC-019, EIA-TC-034, EIA-TC-040, EIA-TC-041, EIA-TC-044, EIA-TC-045, EIA-TC-047, EIA-TC-057
State Transition Testing (ST)	6	EIA-TC-013, EIA-TC-016, EIA-TC-031, EIA-TC-062 to EIA-TC-064
Use Case Testing (UC)	10	EIA-TC-029, EIA-TC-032, EIA-TC-042, EIA-TC-061, EIA-TC-065 to EIA-TC-068, EIA-TC-070, EIA-TC-071

5.5 Execution Metrics and Requirement Coverage Summary

Because this is black box testing, coverage is reported against requirements and scenarios, not against code. The initial figures are shown below; update them after execution.

Table 5.4: Execution Summary

Metric	Value
Total scenarios	15
Total test cases	71
Functional requirements covered	14 of 14 (FR-14 by Blocked cases only)
Test cases executed	[fill after execution]
Passed	[fill]
Failed	[fill]
Blocked	4 (EIA-TC-065 to EIA-TC-068: calculation engine, recommendations and reports not implemented)
Pass percentage	[fill]

5.6 Defect Tracking Register

Record every failed test case here. Defects can also be logged in JIRA and the issue key copied into this table.

Table 5.5: Defect Register

Defect ID	Test case ID	Summary	Severity	Priority	Status
DEF-001					
DEF-002					
DEF-003					
DEF-004					
DEF-005					

5.7 AI Tool Usage and Validation Sign-Off

The same prompt was given to both tools. It asked for functional test scenarios and test cases for the platform, in the standard template, using EP, BVA, Decision Table, State Transition and Use Case techniques. Screenshots of both tool sessions are attached as evidence in the appendix.

Table 5.6: AI Tool Comparison

Criteria	Claude	[Second tool]
Test cases generated	71 (after team review)	62
Techniques covered	EP, BVA, DT, ST, UC	EP, BVA, DT, ST, UC (as requested in the prompt)
Use of project-specific details (radius, providers, coordinates, statuses)	Yes	Yes (password, coordinate, radius and 30-day rules; provider errors; GIS)
Cases that needed correction	[fill]	[fill]
Cases missed by the tool	[fill]	[fill]
Overall remark	[fill]	[fill]

Validation sign-off: the test cases were reviewed by the team and checked against the application requirements before execution.

Team signatures: _____ _____ _____ _____

CHAPTER 6: PLATFORM EVALUATION & APPLICATIONS

6.1 Technical Advantages

- **One workflow for the baseline:** air quality, weather, climate, terrain, soil, land use, ecology, receptors, population and hazards are collected in one request instead of from a dozen separate portals.
- **No invented values:** every provider and every catalogue item reports available, unavailable, error, skipped or project input, with a reason, so gaps are visible.
- **Fault isolation:** providers run concurrently with their own time limit, so one slow or failing provider does not stop the others.
- **Labelled data:** observed, modelled, reanalysis and derived values are marked, and CPCB AQI sub-indices are never mixed with concentrations.
- **Traceable reference data:** each regulatory limit carries a verified flag, and unverified limits are counted on the dashboard.
- **Clear service boundaries:** the stateless analytics service never writes to the database, so storage stays consistent and can be tested separately.

6.2 Structural Limitations and Bottlenecks

- **External availability:** public providers are not always reachable. The main Overpass instance often returns 504 and the CPCB API can return 502, so results vary between runs; tests must check how failures are reported, not only the happy path.
- **Fetch duration:** a full baseline fetch takes 1 to 4 minutes, which limits how often UI tests can be repeated.
- **Coverage gaps:** surface water, groundwater, eco-sensitive zones and official flood or seismic zonation have no integrated dataset; protected areas come from OpenStreetMap, which is context, not regulatory evidence.
- **Roles not enforced:** the four roles are stored but grant the same permissions; access depends only on ownership.
- **Engine not implemented:** impact calculation, compliance, recommendations and reports are not built, so EIA-TC-065 to EIA-TC-068 are Blocked.
- **Unverified reference data:** collected limits and coefficients still await sign-off and must not be used for real results.
- **Assumed rules:** a few expected results marked (A) describe intended behaviour and must be confirmed with the project team.

6.3 Real-World Target Applications

- **Environmental consultants:** a quick, documented baseline for Form 1 and Terms of Reference preparation.
- **Project developers:** early screening of candidate sites in Solapur district for sensitive receptors and data gaps.
- **Regulators and pollution control boards:** a consistent first view of a site's surroundings before detailed review.

- **Academic and training use:** teaching how environmental data sources, GIS distances and regulatory limits fit together in an EIA.

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Conclusion

This activity produced 15 test scenarios and 71 test cases for the AI-Assisted EIA Platform, covering all 14 functional requirements. The five black box techniques were applied where they fit the rules and workflows, and traceability to requirements is shown in the RTM. The input rules were taken from the current build rather than assumed, so the expected results describe what the platform should actually do. AI tools were used to speed up test design, and their output was reviewed before inclusion. The cases are ready to be executed, and the Actual Result, Status and defect register are completed during execution.

7.2 Future Scope

- Run the Blocked cases (EIA-TC-065 to EIA-TC-068) and add cases for impact calculation, compliance, recommendations and reports when the calculation engine is built.
- Automate the stable API cases (registration, login, tokens, validation, ownership) with Postman/Newman or Supertest and run them in CI.
- Add role-based permission tests if role-specific access is introduced.
- Add contract tests with recorded provider responses so the data-quality cases can run without live providers.
- Add performance tests for baseline fetch duration and concurrent users.

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APPENDIX A: AI TOOL SESSION EVIDENCE

Figure A.1 shows the Claude session (15 October 2026) that produced the test cases. Figures A.2 and A.3 show the prompt given to the second AI tool and an extract of the 62 test cases it generated.

Claude Code (VS Code extension) · project: eia-ai-platform — Session: 15 October 2026

USER

take this pdf as refernece and create the same for our project

STQA_Testing_Report_FreshRescue_Revised.pdf

CLAUDE

```
Read backend/src/routes, controllers, validators, middleware
Read analytics-service/app/schemas.py, services/catalogue.py, mappers
Read frontend/src/pages/SignIn.tsx, CreateProject.tsx, Location.tsx
```

The report follows the reference PDF's structure: **15 test scenarios** and **71 test cases** (EIA-TC-001 to EIA-TC-071), traced to 14 functional requirements, using EP, BVA, Decision Table, State Transition and Use Case techniques. The input rules, error messages and provider behaviour come from the actual code. The 4 cases for the calculation engine, recommendations and reports are marked Blocked because they are not built yet.

Sample of the generated test cases:

Test Case ID	Scenario	Description	Test Data	Expected Result	Technique
EIA-TC-002	TS_01	Verify password length lower boundary (minimum 8 characters)	Passwords of 7, 8 and 9 characters: Abc@123, Abcd@123, Abcde@123 (new email each)	7 characters rejected: UI blocks submit, API returns 400 'Password must be at least 8 characters.'; 8 and 9 characters accepted	BVA
EIA-TC-017	TS_03	Verify a user cannot read, change or delete another user's project	Project ID of P	404 'Project not found.' for each request; P unchanged for A; UI shows an error state	DT
EIA-TC-024	TS_04	Verify latitude boundaries (-90 to 90)	Latitude: -90.0001, -90, 90, 90.0001 (Longitude 75.9064)	-90.0001 and 90.0001 rejected (UI: 'Latitude must be between -90 and 90.'; API: 400); -90 and 90 accepted	BVA
EIA-TC-044	TS_08	Verify a provider without credentials is skipped	Same site and radius	openaq status 'skipped' with a reason; other providers unaffected; response still 201	DT
EIA-TC-054	TS_10	Verify absence is reported explicitly	Site with no protected area within 1 km; radius 1 km	Reported as 'none found within 1 km'; item status available; no distance invented	EP

Excerpt of the session transcript; the full set of cases is in Section 5.2.

Figure A.1: Claude session – request and generated test cases (excerpt)

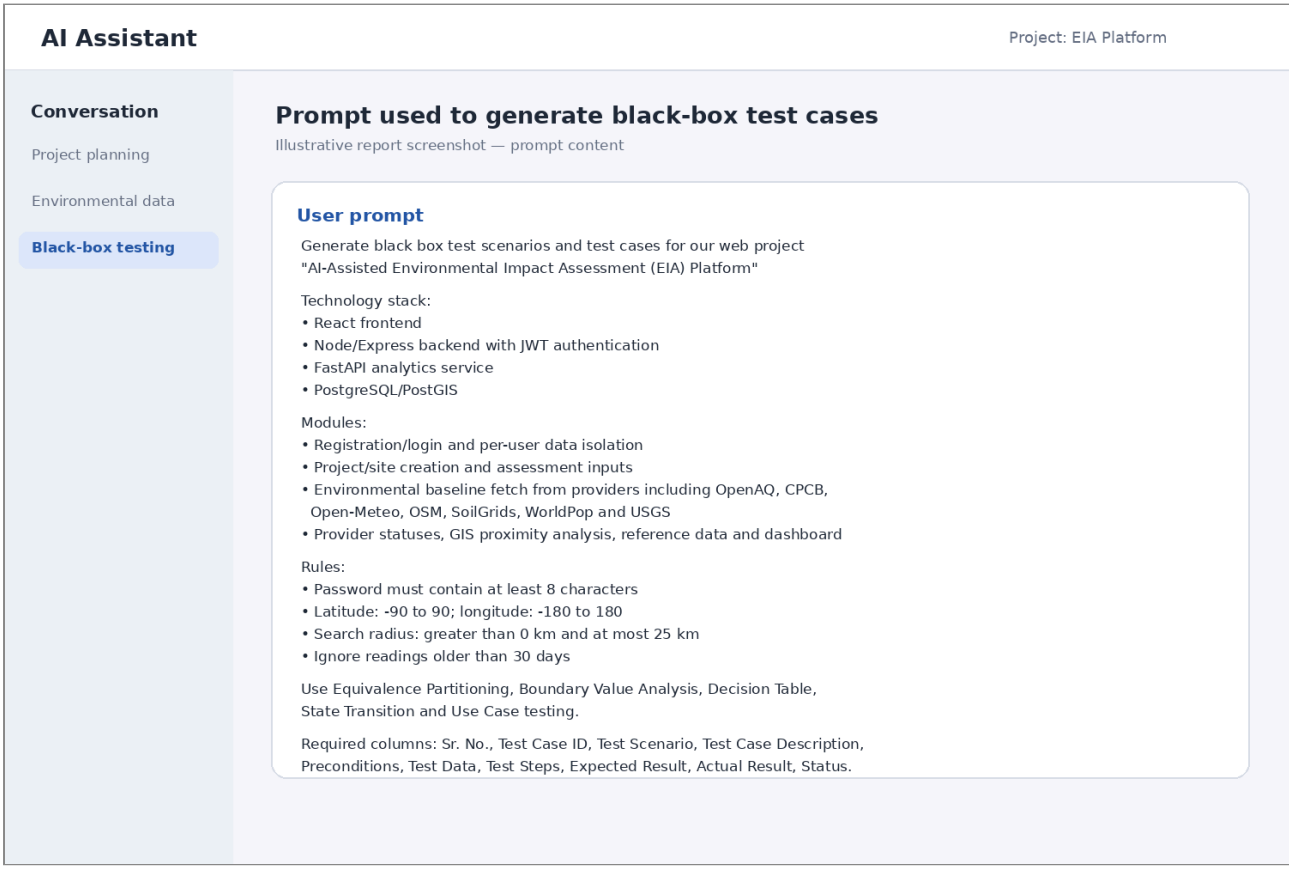


Figure A.2: Prompt given to the second AI tool

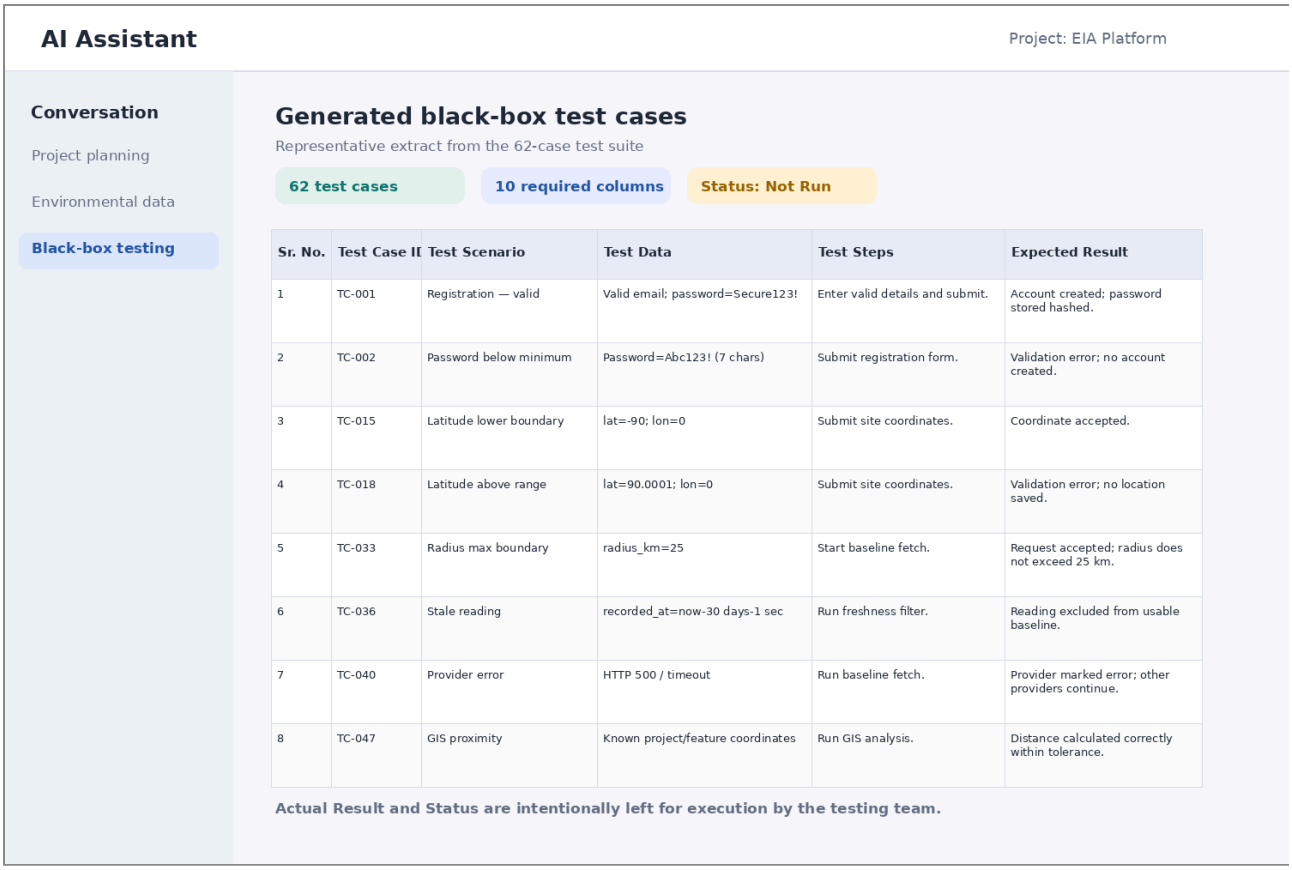


Figure A.3: Test cases generated by the second AI tool (extract of 62 cases)